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TrustCore NX Architecture – Quantum-Secure RISC-V SoC Platform

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Title

Quantum-Secure RISC-V SoC with Integrated Post-Quantum Crypto Accelerators and Hardware Root-of-Trust

Technical Field

The invention relates to secure embedded computing systems, in particular to system-on-chip (SoC) architectures combining RISC-V vector processing with post-quantum cryptographic (PQC) acceleration and trusted platform hardware security modules.

Background

Existing SoCs implement security mainly through software-based cryptographic libraries or external TPM modules. Such configurations are vulnerable to side-channel attacks, latency, and power inefficiencies. As post-quantum algorithms such as Kyber and Dilithium become mandatory for secure communication, the lack of hardware-level PQC integration presents a critical bottleneck for edge and defense applications.

Summary of the Invention

The invention provides a unified SoC design, TrustCore NX, in which post-quantum cryptographic accelerators are embedded directly into RISC-V vector lanes (RVV 1.0). The design integrates TPM 2.0 key storage, AES-512 encryption engines, and a 32 TOPS neural co-processor, enabling simultaneous secure computation and AI-driven sensor fusion. This combination creates a hardware root-of-trust capable of quantum-resistant operation and real-time autonomy at the edge.

Detailed Description

TrustCore NX comprises a 12-core RISC-V cluster connected to an internal security bus. Each vector core includes microcoded PQC instructions enabling Kyber and Dilithium operations in parallel. The architecture integrates the following components:

- RISC-V RVV Vector Cluster (12 cores)
- TPM 2.0 Secure Key Storage and Random Number Generator
- PQC Hardware Accelerators for Kyber/Dilithium
- AES-512 Engine for symmetric encryption
- Neural Co-Processor (32 TOPS) optimized for sensor fusion and anomaly detection. Power management and telemetry subsystems ensure field operation stability in defense, satellite, and industrial environments.

Claims

1. A system-on-chip (SoC) architecture comprising: a plurality of RISC-V vector cores configured for parallel cryptographic processing; hardware modules implementing Kyber and Dilithium post-quantum algorithms; an integrated Trusted Platform Module (TPM 2.0) for key storage; wherein the SoC provides post-quantum secure communication and neural computation within a single silicon package.
2. The system of claim 1, wherein the post-quantum accelerators are implemented within RISC-V vector lanes using dedicated microcode instructions.
3. The system of claim 1 or 2, wherein the SoC integrates an AES-512 symmetric encryption engine and a 32 TOPS neural coprocessor for secure AI operations.
4. The system of any preceding claim, wherein the TPM and PQC accelerators share a unified internal security bus for real-time key exchange.
5. The system substantially as described herein with reference to the accompanying block diagrams and examples.

Drawings

To be attached: Block Diagram of TrustCore NX SoC, PQC Accelerator Flowchart, and Integration Layout.